

We modeling penalty corners in field hockey as a game between the offense and defense. At each corner, the offense chooses a setup and strategy with the objective of scoring, while the defense chooses a strategy to prevent a goal. Corners are generated by fouls committed inside the circle, so the number of corners in a match is not known in advance. In particular, any corner may be the final corner of the match, but teams do not know *ex ante* when the final corner will occur.

We therefore model corners as an indefinite horizon zero sum repeated game in which each corner constitutes a stage game. Each stage game has two parts: teams first observe the current state of the match, and then simultaneously choose offensive and defensive strategies.

$$\text{For each corner } t : \left\{ \begin{array}{l} S_t = (H_t, M_t, F_t^O, F_t^D), \\ \sigma^O(\cdot | S_t) \in \Delta(\mathcal{A}^O), \quad \sigma^D(\cdot | S_t) \in \Delta(\mathcal{A}^D), \\ A_t^O \sim \sigma^O(\cdot | S_t), \quad A_t^D \sim \sigma^D(\cdot | S_t), \\ Y_t \in \{\text{Goal}, \text{No Goal}\}, \\ H_{t+1} = (H_t, A_t^O, A_t^D, Y_t), \\ S_{t+1} = (H_{t+1}, M_{t+1}, F_{t+1}^O, F_{t+1}^D). \end{array} \right.$$

where

- S_t is the state of the game at corner t :
 - H_t : corner history (past actions, setups, and outcomes; potentially weighted toward the current game—see below).
 - M_t : match context, comprising `quartert` (match quarter), `score_statet` (attacking team's relative score), `corner_numbert` (attacking team's corner sequence in the match), Elo_t^O (attacking team pre-game Elo), and Elo_t^D (defending team pre-game Elo).
 - F_t^O : offensive setup, comprising `trapt` (trapping option) and `zonet` (injection/shot zone). Observed by both teams before strategies are chosen.
 - F_t^D : initial formation. Observed by both teams before strategies are chosen.
- $\mathcal{A}^O$ and $\mathcal{A}^D$ are the sets of possible offensive and defensive actions, respectively.
- $\sigma^O(\cdot | S_t)$ and $\sigma^D(\cdot | S_t)$ are the offensive and defensive strategies conditional on the state S_t . Each strategy specifies a probability distribution over the relevant action set.
- A_t^O and A_t^D are the realized offensive and defensive actions drawn from these strategies.
- Y_t is the realized outcome of the corner.
- S_{t+1} is the state at the next corner. Thus, after each corner, the state is updated to incorporate the new information generated by the current corner.

Stage Game

Part 1: Common information is revealed. Once teams know that a penalty corner will occur, they observe the match context (quarter, score state, corner number, team Elo ratings), the offensive setup (trap, zone), and the defensive initial formation.

Part 2: Strategies are chosen simultaneously. Conditional on this information, the offense and defense simultaneously choose their strategies. The resulting outcome is then realized (e.g., goal or no goal).

We know that a mixed strategy Nash equilibrium exists in this game for each state S_t by Nash's existence theorem. Thus we can calculate best response correspondences for each team.

Proof of Concept/Simple Example:

Consider some state S_t .

The offense has strategies: $\sigma_O(\cdot \mid S_t)$ where their action A_t^O will be drawn from.

The defense has strategies: $\sigma_D(\cdot \mid S_t)$ where their action A_t^D is drawn from.

The offense maximizes:

$$u^O(A_t^O, A_t^D, S_t) = P(Y_t = \text{Goal} \mid A_t^O, A_t^D, S_t)$$

Since the game is zero sum, the defense gets:

$$u^D(A_t^O, A_t^D, S_t) = -u^O$$

Using existing data, we have some payoff matrix that we can populate for each S_t :

	A_1^O	A_2^O	A_3^O	$\dots$	A_n^O
A_1^D					
A_2^D					
A_3^D					
$\vdots$					
A_h^D					

This is what we need to calculate with data. Each entry is just

$$P(\text{Goal} \mid A_t^D, A_t^O, S_t).$$

For example, with two strategies, we might have the following payoff matrix:

	A_1^O	A_2^O
A_1^D	0.5	0.5
A_2^D	0.75	0.25

which would represent the following game, where offense maximizes the probability of a goal and defense minimizes it. Equivalently, if defense utility is defined as the negative probability of a goal, both players maximize their own utility:

	A_1^O	A_2^O
A_1^D	(−.5, .5)	(−.5, .5)
A_2^D	(−.75, .75)	(−.25, .25)

The upper-left cell is a Nash equilibrium. Given that the offense plays A_1^O , the defense prefers A_1^D , since

$$-.5 > -.75.$$

Given that the defense plays A_1^D , the offense is indifferent between A_1^O and A_2^O , since both give payoff .5. Thus neither player has a profitable unilateral deviation from (A_1^D, A_1^O) .

We could also have a game like this:

	A_1^O	A_2^O
A_1^D	(−.25, .25)	(−.75, .75)
A_2^D	(−.75, .75)	(−.25, .25)

Here, there is no pure strategy Nash equilibrium.

Let $p = \Pr(A_1^O)$ and $q = \Pr(A_1^D)$. We look for indifference point:

$$0.75p + 0.25(1 - p) = .25p + 0.75(1 - p)$$

$$p = 1/2$$

and

$$0.25q + 0.75(1 - q) = 0.75q + 0.25(1 - q)$$

$$q = 1/2$$

Thus the Nash is $p = 1/2, q = 1/2$.

When we have a larger matrix, we can eliminate any dominated strategies and then solve for a Nash equilibrium. The resulting equilibrium will specify the probability with which each team should play each strategy in its equilibrium support.

Weighted Estimation of Payoff Matrices:

To populate each cell of the payoff matrix, we estimate $P(\text{Goal} \mid A^O, A^D, S_t)$ from observed corners with matching action profiles and similar states. Rather than weighting all observations equally, we can weight corners from the game of interest more heavily than corners from other games. This reflects the idea that same-game observations—against the current opponent, in the current match context—are more informative about the relevant strategic environment.

Define weights for each historical corner τ :

$$w_\tau = \begin{cases} \lambda & \text{if corner } \tau \text{ is from the current game,} \\ 1 & \text{otherwise,} \end{cases}$$

where $\lambda > 1$. The weighted estimate of each payoff matrix entry is then

$$\hat{P}_w(\text{Goal} \mid a^O, a^D, s) = \frac{\sum_{\tau} w_{\tau} \cdot \mathbf{1}(Y_{\tau} = \text{Goal}, A_{\tau}^O = a^O, A_{\tau}^D = a^D)}{\sum_{\tau} w_{\tau} \cdot \mathbf{1}(A_{\tau}^O = a^O, A_{\tau}^D = a^D)}$$

where the sums range over corners τ with states S_{τ} in the relevant conditioning set. When $\lambda = 1$, all games are weighted equally. As λ increases, the estimates tilt toward same-game frequencies, making the equilibrium analysis more opponent-specific.

This nests two extremes: using only league-wide data ($\lambda = 1$) and using only same-game data ($\lambda \rightarrow \infty$). The choice of λ trades off bias (from pooling dissimilar opponents) against variance (from small same-game samples), and can be selected by cross-validation or treated as a tuning parameter.

Results:

For now, we do not condition on the corner history H_t . The payoff matrix below is estimated by pooling all corners.

Some action pairs are rarely observed. For example, “back to inserter” against “2-2 to 3-1” appears in only 3 corners, none of which were goals, so the raw goal rate for that cell is 0%. The linear program takes such estimates at face value and would treat that pair as a

guaranteed stop. To avoid this, we smooth each cell with a Laplace prior,

$$\hat{P}(\text{Goal} \mid a^D, a^O) = \frac{g(a^D, a^O) + \alpha}{n(a^D, a^O) + 2\alpha},$$

where g and n are the number of goals and corners in the cell. This is equivalent to adding α goals and α non-goals to every cell before computing the rate, so every cell is pulled toward $1/2$. We use $\alpha = 0.2$, a weak prior that adds less than half a corner to each cell. Sparse cells move up from 0%: the cell above becomes $0.2/3.4 = 5.9\%$. Meanwhile well-populated cells are essentially unchanged (“hit” against “3-1 to 3-1”, with 2,170 corners, stays at 14.8%). The equilibrium is stable for values of α up to about 0.35: the same actions receive positive weight and goal rates change smoothly. Results are in Table 1.

The following table shows the raw strategy we see in the data, what would happen if the offense and defense or both of them played a Nash equilibrium, and the resulting goal rates. The Nash equilibrium is computed from the smoothed payoff matrix.

	Observed	Nash offense only	Nash defense only	Nash equilibrium
Offense σ^O	Sweep 27.0% Hit 26.3% Flick 21.3% Pass/stick 13.1% Pass/glove 8.5% BTI 3.2% OSI 0.6%	Flick 67.3% Hit 26.8% Pass/glove 5.9%	(unchanged)	Flick 67.3% Hit 26.8% Pass/glove 5.9%
Defense σ^D	3-1 to 3-1 90.8% 3-1 to 2-2 3.8% 3-1 reverse 2.4% 2-2 to 3-1 1.4% 2-2 to 2-2 0.8% 3 man defense 0.4% Double runner 0.3%	(unchanged)	2-2 to 3-1 71.5% 3-1 to 2-2 27.6% Double runner 0.8%	2-2 to 3-1 71.5% 3-1 to 2-2 27.6% Double runner 0.8%
Goal Rate	16.5%	17.7% (+7.4%)	12.8% (−22.2%)	16.3% (−0.9%)

Table 1: Payoff matrix entries smoothed via Laplace prior ($\alpha = 0.2$). Percentage changes are relative to observed goal rate.